

PR4MA1104 MATEMATIKA I

No. _____

Date: _____

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Jawaban

1. $\lim_{x \rightarrow 1} \frac{\sqrt{2x} |x-1|}{x-1} =$ Tidak ada nilai limitnya.
 $x > 1 \Rightarrow |x-1| = x-1$

Bukti. Tinjau $\lim_{x \rightarrow 1^+} \frac{\sqrt{2x} |x-1|}{x-1} = \frac{\sqrt{2x}(x-1)}{x-1}$

$$= \sqrt{2x}$$

$$= \sqrt{2} \lim_{x \rightarrow 1^+} x$$

$$= \sqrt{2}$$

$\checkmark x < 1 \Rightarrow |x-1| = -(x-1)$

Tinjau $\lim_{x \rightarrow 1^-} \frac{\sqrt{2x} |x-1|}{x-1} = \frac{\sqrt{2x} (-(x-1))}{x-1}$

$$= -\sqrt{2x}$$

$$= -\sqrt{2}$$

Karena $\lim_{x \rightarrow 1^+} \frac{\sqrt{2x} |x-1|}{x-1} \neq \lim_{x \rightarrow 1^-} \frac{\sqrt{2x} |x-1|}{x-1}$,

nilai limitnya tidak ada!

2. $\lim_{x \rightarrow 0} \frac{\tan^2 3x}{2x} = \lim_{x \rightarrow 0} \frac{\tan 3x \cdot \tan 3x}{2x}$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{\cos 3x} \cdot \frac{\sin 3x}{\cos 3x} \cdot \frac{1}{2x}$$

$\hookrightarrow \lim_{x \rightarrow 0} x \cdot x$
 $\Rightarrow \frac{(3x)^2}{2x}$
 $= \frac{9x}{2}$
 $= \frac{9}{2}x$
 $= 0$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{3x}{2x} \cdot \frac{\sin 3x}{\cos 3x} \cdot \frac{1}{1} \cdot \frac{1}{1}$$

$$= \lim_{x \rightarrow 0} 1 \cdot \frac{3}{2} \cdot 0 = \boxed{0}$$

$$\begin{aligned}
 3. \quad a. \lim_{x \rightarrow -\infty} \frac{\sqrt{1+4x^6}}{2-x^3} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{1+4x^6} \cdot \frac{1}{x^3}}{(2-x^3) \cdot \frac{1}{x^3}} \\
 &= \lim_{x \rightarrow -\infty} \frac{\sqrt{\frac{1+4x^6}{x^6}}}{\frac{2}{x^3}-1} \\
 &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^{-6}+4}}{2x^{-3}-1} = \frac{\sqrt{0+4}}{2(0)-1} \\
 &= \frac{\sqrt{4}}{1} = \frac{2}{1} = 2
 \end{aligned}$$

$$\begin{aligned}
 b. \lim_{x \rightarrow \infty} \sqrt{9x^2+x} - 3x \\
 &= \lim_{x \rightarrow \infty} \sqrt{9x^2+x} - \sqrt{9x^2} \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt{9x^2+x} - \sqrt{9x^2} \cdot (\sqrt{9x^2+x} + \sqrt{9x^2})}{1 \cdot (\sqrt{9x^2+x} + \sqrt{9x^2})} \\
 &= \lim_{x \rightarrow \infty} \frac{9x^2+x - (9x^2)}{\sqrt{9x^2+x} + \sqrt{9x^2}} \\
 &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{9x^2+x} + \sqrt{9x^2}} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \\
 &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{9+\frac{1}{x}} + \sqrt{9}} \\
 &= \frac{1}{\sqrt{9+0} + \sqrt{9}} = \frac{1}{3+3} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 c. \lim_{x \rightarrow -\infty} \frac{1+x^6}{x^4+1} &= \lim_{x \rightarrow -\infty} \frac{(1+x^6)(x^{-4})}{(x^4+1)(x^{-4})} \\
 &= \lim_{x \rightarrow -\infty} \frac{x^{-4}+x^2}{1+\frac{1}{x^4}} \\
 &= (-\infty)^2 \\
 &= \infty^2 = \boxed{\infty} \text{ (diverge)}
 \end{aligned}$$

(part seru)

Mari kita buktikan bahwa $\lim_{x \rightarrow -\infty} \frac{1+x^6}{x^9+1} = -\infty$

menggunakan definisi limit untuk $-\infty \xrightarrow{\text{mendekati}} \infty$

Bukti. Misalkan $\exists N > 0 \Rightarrow \exists M \ x < M, \frac{1+x^6}{x^9+1} > N$

Ambil $M = \min(-1, -\sqrt{2N})$

karena $x < M$, $x < -1 \wedge x < -\sqrt{2N}$

karena $x < -1$, $x^9 > 1 \Rightarrow x^9 + 1 < 2x^9$

$$1+x^6 > x^6 \Rightarrow \frac{1+x^6}{x^9+1} > \frac{x^6}{x^9+1} > \frac{x^6}{2x^9} > \frac{x^2}{2} > N$$

Maka, $\exists N > 0, \frac{1+x^6}{x^9+1} > N$

Q.E.D.

d. $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = \boxed{0}$

↳ Squeeze theorem/sandwich

Bukti 1. Perhatikan bahwa $-1 \leq \sin x \leq 1$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

$$\Rightarrow 0 \leq \frac{\sin x}{x} \leq 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$$

Bukti 2. Menggunakan definisi limit untuk tak hingga

$$\forall \epsilon > 0, \exists M < x \Rightarrow \left| \frac{\sin x}{x} - 0 \right| < \epsilon$$

Ambil $M = \frac{1}{\epsilon}$, maka karena $|\sin x| \leq 1$,

$$\left| \frac{\sin x}{x} \right| \leq \frac{1}{|x|} \leq \frac{1}{x} < \epsilon$$

$$\hookrightarrow x > M > 0 \Rightarrow |x| = x$$

9.

$$a. \lim_{x \rightarrow 2^-} \frac{x^2}{4-x^2} = \boxed{\infty}$$

Bukti 1 Substitusi x dengan $x = 2-h$ dengan $h > 0$,
maka ekuivalen dengan

$$\lim_{h \rightarrow 0^+} \frac{(2-h)^2}{4-(2-h)^2} = \frac{4+h^2-4h}{4-(4+h^2-4h)}$$

$$= \frac{4+h^2-4h}{4h-h^2} \cdot \frac{\frac{1}{h^2}}{\frac{1}{h^2}}$$

$$= \frac{\frac{4}{h^2} + 1 - \frac{4}{h}}{\frac{1}{h} - 1}$$

Substitusi $h=0$

$$\Rightarrow \frac{4+(0)^2-4(0)}{4(0)-0^2} = \frac{4}{0^+} = \boxed{\infty}$$

Bukti 2 Menggunakan definisi limit, maka
 $\forall N > 0, \exists \delta > 0 \mid 2-\delta < x < 2 \Rightarrow \frac{x^2}{4-x^2} > N$

Ambil $\delta = \min(1, \frac{1}{4N})$, maka

$$2-\delta < x < 2 \Rightarrow x > 1 \wedge 2-x < \frac{1}{4N}$$

$$\frac{x^2}{4-x^2} > \frac{x^2}{(2-x)(2+x)} > \frac{1}{4(2-x)} > \frac{1}{4 \cdot \frac{1}{4N}} = N$$

$(x \in (1, 2))$

$$b. \lim_{t \rightarrow \pi^-} \cot t = \boxed{-\infty}$$

Bukti. $\forall N < 0, \exists \delta > 0 \mid \pi - \delta < t < \pi, \cot(t) < N$

Perhatikan bahwa $t > t_0 \Rightarrow \cot(t) < \cot(t_0)$

Ambil $\delta = \pi - \operatorname{arccot}(N)$. Karena $\cot$ kontinu,
 arccot juga kontinu Sehingga

$$t > \pi - \delta \Rightarrow t > \pi - (\pi - \operatorname{arccot}(N))$$

$$\Rightarrow t > \operatorname{arccot}(N)$$

$$\Rightarrow \cot t < N$$

5. Asimtot datar/tegak dari $f(x) = \frac{1+x^4}{x^2-x^4}$

Asimtot tegak = pembilang nol
datar = $\lim_{x \rightarrow \infty}$ atau $-\infty$

$$f(x) = \frac{1+x^4}{x^2-x^4} = \frac{1+x^4}{x^2(1-x^2)}, \text{ penyebut nol ketika } x^2(1-x^2)=0$$

$$\Rightarrow x \in \{0, -1, 1\}$$

$$\lim_{x \rightarrow \infty} \frac{1+x^4}{x^2-x^4} = \lim_{x \rightarrow \infty} \frac{1+x^4}{x^2-x^4} = -1 \quad \therefore \text{fungsi genap, } f(x) = f(-x)$$

Dengan demikian, ada

3 asimtot tegak pada $x=0$, $x=-1$, dan $x=1$;
1 asimtot tegak pada $y=-1$

6. $f(x) = \frac{x^2+3x-10}{x^2-5x+6} = \frac{(x+5)(x-2)}{(x-3)(x-2)}$

Ketidakkontinuan terdapat pada $x=3$ atau $x=2$ karena pembuat penyebut nol dengan $x=3$ faktorthapuskan, Sementara $x=2$ terhapuskan karena

$$\lim_{x \rightarrow 2} \frac{(x+5)(x-2)}{(x-3)(x-2)} = \lim_{x \rightarrow 2} \frac{x+5}{x-3} = \frac{7}{-1} = -7$$

$x=3$ faktorthapuskan karena berupa asimtot tegak dan nilai

$$\lim_{x \rightarrow 3} \frac{(x+5)(x-2)}{(x-3)(x-2)} \quad \text{+1, tidak ada}$$

$$f(x) = \begin{cases} cx^2 + 2x, & x < 2 \\ x^3 - cx, & x \geq 2 \end{cases}$$

Agar kontinu, $\lim_{x \rightarrow 2} f(x)$ harus ada, maka

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\Rightarrow \lim_{x \rightarrow 2^-} cx^2 + 2x = \lim_{x \rightarrow 2^+} x^3 - cx$$

$$\Rightarrow 4c + 2c = 8 - 2c$$

$$\Rightarrow 8c = 8$$

$$\Rightarrow \frac{8c}{8} = \frac{8}{8}$$

8. Bukti bahwa $\exists x \in \mathbb{R} \Rightarrow \sqrt{x-5} = \frac{1}{x+3}$ dapat menggunakan

IVT atau Teorema Nilai Tengah / Antara
konstruksi sebuah fungsi $h(x) = \sqrt{x-5} - \frac{1}{x+3}$

$$h(5) = \sqrt{5-5} - \frac{1}{5+3} = -\frac{1}{8} < 0$$

$$h(6) = \sqrt{6-5} - \frac{1}{6+3} = 1 - \frac{1}{9} = \frac{8}{9} > 0$$

Karena h kontinu pada $x > 5$, $\exists c \Rightarrow -\frac{1}{8} < 0 < \frac{8}{9}$
dan karena $0 \in [-\frac{1}{8}, \frac{8}{9}]$, maka
 $5 < c < 6$.

$$h(5) < h(c) < h(6)$$

Dengan demikian $\exists c \in (5, 6) \Rightarrow h(c) = 0$

karena $0 \in (-\frac{1}{8}, \frac{8}{9})$